

Logical Operators Quick Reference				
$\neg$	$\wedge$	$\vee$	$\rightarrow$	$\leftrightarrow$
$\forall$	$\exists$	$\in$	$\notin$	$\equiv$

1A. $f(x, y) = x^2 + y + 1$ is **onto but not one-to-one** since $f(-1, 1) = f(1, 1) = 3$.

1B. $f(x, y) = y! - 3x$ is **onto but not one-to-one** since $f(1, 1) = f(1, 0) = -2$.

1C. $f(x, y) = 9y^2 + 3y + 2$ is **neither onto nor one-to-one** since $f(0, 1) = f(e, 1) = 14$ and $f(x, y) \neq 0$.

1D. $f(x, y) = \frac{x}{y} + y$ is **neither onto nor one-to-one** since $f(-6, 3) = f(0, 1) = 1$ and $f(x, y) \neq 0$.

1E. $f(x, y) = 30x + |y|$ is **onto but not one-to-one** since $f(0, -1) = f(0, 1) = 1$.

For problem two, let the following array represent the change:
[quarters, dimes, nickels, pennies].

2A. $[1, 1, 1, 4] = 1(0.25) + 1(0.10) + 1(0.05) + 4(0.01) = \0.44

2B. $[2, 2, 0, 4] = 2(0.25) + 2(0.10) + 0(0.05) + 4(0.01) = \0.74

2C. $[3, 1, 1, 3] = 3(0.25) + 1(0.10) + 1(0.05) + 3(0.01) = \0.93

3A. We will prove the statement using a total amount of money of \$0.19. Using the Greedy Cashier's Algorithm without the \$0.14 coin, the produced array is:

$[0, 1, 1, 4] = 0(0.25) + 1(0.10) + 1(0.05) + 4(0.01) = \0.19 for a total of 6 coins

However, if the \$0.14 coin is used – by use of the array format [quarters, fourteen-cent, dimes, nickels, pennies] – the produced array is:

$[0, 1, 0, 1, 0] = 0(0.25) + 1(0.14) + 0(0.10) + 1(0.05) + 0(0.01) = \0.14 for a total of 2 coins.

Therefore, we have found an example of the statement being proven since using the fourteen-cent coin *can* use fewer coins since $2 < 6$.

3B. We will disprove the statement using a total amount of money of \$0.20. Using the Greedy Cashier's Algorithm without the \$0.14 coin, the produced array is:

$[0, 2, 0, 0] = 0(0.25) + 2(0.10) + 0(0.05) + 0(0.01) = \0.20 for a total of 2 coins

However, if the \$0.14 coin is used – by use of the array format [quarters, fourteen-cent, dimes, nickels, pennies] – the produced array is:

$[0, 1, 0, 1, 1] = 0(0.25) + 1(0.14) + 0(0.10) + 1(0.05) + 1(0.01) = \0.20 for a total of 3 coins.

Therefore, we have found an example of the statement being disproven since using the fourteen-cent coin *will not* necessarily produce change in the fewest coins possible for all coins since $2 < 3$.

4A. The statement is true. Since x and c are positive integers and $x + c < 5$, it holds true (by division of both sides by 5) that $\frac{x+c}{5} < 1$. Since $\frac{x+c}{5}$ must always be a positive fraction that is less than one, it holds true that $\left\lfloor \frac{x+c}{5} \right\rfloor = 0$. Similarly, we know x attains its maximum value when bounded by $x + c < 5$ when $c = 0$ – assuming 0 is a positive integer (if 0 is not a positive integer, we know for $c = 1$ the proof holds since $1 > 0$). Thus, we have $x < 5$. Since we know the maximum value of x is positive and less than 5, we know $\frac{x}{5} < 1$ by division of 5 on both sides. Since $\frac{x}{5}$ must always be a positive fraction that is less than one, it holds true that $\left\lfloor \frac{x}{5} \right\rfloor = 0$. Therefore, we have proved the equality of $\left\lfloor \frac{x+c}{5} \right\rfloor = \left\lfloor \frac{x}{5} \right\rfloor = 0$ and so the statement is true.

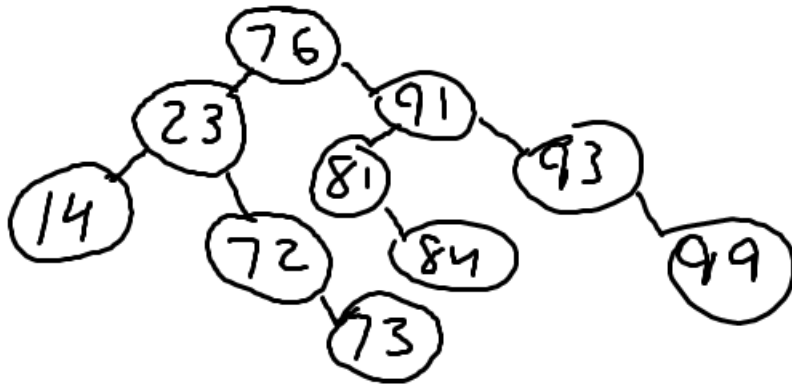
4B. The statement is false. Let $g(x) = \arcsin(x)$ and $f(x) = \frac{1}{x}$. We know $g(f(x)) = \arcsin\left(\frac{1}{x}\right)$ and $f(g(x)) = \frac{1}{\arcsin(x)}$. Let $x = 2$; we know $g(f(2)) = \arcsin\left(\frac{1}{2}\right)$ is defined since the range of arcsine is $[-1, 1]$. However, for the same reason, $f(g(x)) = \frac{1}{\arcsin(x)}$ is undefined since $\arcsin(2)$ is undefined. Therefore, we have shown that if $f(g(x))$ is undefined, then $g(f(x))$ is not necessarily undefined.

5A. $f(x)$ is $O(g(x))$ and $g(x)$ is $O(f(x))$ since both $f(x)$ and $g(x)$'s component of greatest eventual growth is x .

5B. $f(x)$ is $O(g(x))$ since the component of greatest eventual growth is greater for $g(x)$ than for $f(x)$.

5C. $f(x)$ is $O(g(x))$ since the component of greatest eventual growth is greater for $g(x)$ than for $f(x)$.

5D. $f(x)$ is $O(g(x))$ since the component of greatest eventual growth is greater for $g(x)$ than for $f(x)$.



6. Given the above Balanced Binary Search tree, we will search for 30 in the binary tree. The steps followed are below (were each line is a new node) – while starting at the root:

76 $\neq$ 30 | 76 > 30

23 $\neq$ 30 | 23 < 30

72 $\neq$ 30 | 72 > 30

Null $\neq$ 30

7. We will prove the statement using witness.

$$2x^4 \log(x) \leq x^5 \quad \forall x \geq 0$$

$$x^3 \log(x) \leq x^5 \quad \forall x \geq 0$$

$$3x \log(x) \leq x^5 \quad \forall x \geq 0$$

$$\frac{9}{x \log(x)} \leq x^5 \quad \forall x \geq 2$$

Therefore, we have found that $f = O(x^5)$ for $c = 4$ and $k = 2$ meaning $f \leq 4x^5$ for all $x > 2$. Therefore, we have proven the statement.

8A. We will prove the statement using witness.

$$2x^2 \log(x^3) \leq 3x^3 \quad \forall x \geq 0$$

Therefore, we have found that $f = O(3x^3)$ for $c = 1$ and $k = 0$ meaning $f \leq 4x^5$ for all $x > 0$. Therefore, we have proven the statement.

8B. We will evaluate the statement using witness.

$$3x^3 \not\leq 2x^2 \log(x^3) \quad \forall x$$

Therefore, we have not found that $f = O(3x^3)$ since the equality $g(x)$ is never greater than $f(x)$ and so we cannot find a value c or k to make the statement true. Therefore, we will use proof by contradiction to show why $g(x)$ is not $O(f(x))$.

We will assume $g(x) = O(f(x))$. If this is true, then $\exists k > 0$ and $\exists c > 0$ such that $3x^3 \leq c2x^2 \log(x^3) \quad \forall x \geq k$. We will divide both sides of the expression by x^2 giving us $3x \leq 2c \log(x^3)$. We will isolate c by dividing $2 \log(x^3)$ on both sides. We now have $\frac{3x}{2 \log(x^3)} \leq c$. No matter what c and k are, the inequality $\frac{3x}{2 \log(x^3)} \leq c$ does not hold for $x = \max(k + 1, c + 1)$. Therefore, we arrive at a contradiction. Therefore, by proof of contradiction, since our assumption of there exists a k and there exists a c greater than zero such that the expression is true was proven to be false, we arrive at a contradiction and the statement is proven to be false. Therefore, the original statement is disproven.

9. $O(n \log(n))$

10. $O(\log(n))$